



FUJIAN JIANGLE DAYAN 32MW HYDROPOWER PROJECT



Document Prepared By LGAI Technological Center, S.A. (Applus+ Certification)

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Summary:

LGAI Technological Center, S.A. (hereafter referred to as “Applus+ Certification”) has been commissioned by Beijing 3050 Technology Co.,Ltd to perform the verification of greenhouse gas emission reductions of the project activity “Fujian Jiangle Dayan 32MW Hydropower Project” (VCS Ref. No. 246, hereafter referred to as “the project activity”) reported in the monitoring report /1/ during monitoring period 28/03/2006 to 27/03/2012.

The project activity has been validated by TÜV SÜD Industrie Service GmbH based on the VCS PD /3/ version 2 dated 29/10/2009 and reported in the validation report No. 600500301 /4/, completed on 18/11/2009 which is available at <https://registry.verra.org/app/projectDetail/VCS/246>.

The project activity is a hydropower generation project located at Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China, of which the main purpose is to use hydropower for electricity generation. The total installed capacity of the project activity is 32 MW, consisting of 2 sets of hydropower turbine generators with unit capacity of 16 MW. During the 1st renewable 10 years crediting period, the average annual power delivered to the grid by the project is expected to be 120,282 MWh. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of East China Power Grid (ECPG). It’s estimated that the proposed project could achieve GHG emission reductions of 105,990 tCO₂e annually.

The purpose and scope of this verification is to ensure that reported emission reductions are complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements in order to be certified. A desk review and a site visit have been conducted to verify the data submitted in the monitoring report /1/. Applus+ Certification confirms the following has been reviewed:

- Monitoring plan included in the registered VCS PD /3/ version 2 dated 29/10/2009;

- Validation report No. 600500301 /4/, dated 18/11/2009;
- Approved methodology, ACM0002: Grid-connected electricity generation from renewable sources, version 10.0 /7/;
- VCS standards version 4.7 and VCS program guide version 4.4, as well as relevant UNFCCC requirements;
- All information and references relevant to the project activity's resulting in emission reductions.

During this verification, no finding was identified related to the monitoring, implementation or operations of the project activity in relation to relevant VCS standards, guidance and UNFCCC requirements and relevant host party criteria and the applied baseline and monitoring methodology etc.

Applus+ Certification confirms that the project is implemented in accordance with the registered PD /3/. The monitoring plan complies with the applied methodology ACM0002 version 10.0 /7/ and the monitoring has been carried out in accordance with the registered PD. The monitoring system is in place and the emission reductions are calculated without material misstatements. The level of assurance of the verification is reasonable. Our opinion relates to the projects GHG emissions and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the information reviewed and evaluated Applus+ Certification confirms that the implementation of the project has resulted in 563,855 tCO₂e emission reductions during period 28/03/2006 to 27/03/2012.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center, S.A. (Applus+ Certification) has been commissioned by Beijing 3050 Technology Co.,Ltd to perform the verification of greenhouse gas emission reductions of the project activity “Fujian Jiangle Dayan 32MW Hydropower Project” (VCS Ref. No. 246) reported in the Monitoring Report /1/ during monitoring period 28/03/2006 to 27/03/2012.

LGAI Technological Center, S.A. (Applus+ Certification) as the verification body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065:2020.

The objective of verification is to have an independent review and ex post determination by a Validation and Verification Body (VVB) of the monitored reductions in GHG emissions that have occurred as a result of the registration of VCS project. Certification is the written assurance by the VVB that, during a specific time period, a proposed VCS project activity achieved the reductions in anthropogenic emissions by sources of GHGs as verified.

The objective of this verification/certification is to verify and certify emission reductions, reported for the “Fujian Jiangle Dayan 32MW Hydropower Project” in China for the period 28/03/2006 to 27/03/2012.

1.2 Scope and Criteria

The verification scope is defined as an independent and objective review of the registered PD, the Project’s baseline study and Monitoring Report (MR) and other relevant documents. The information in these documents is reviewed against VCS Version 4.7 requirements, UNFCCC rules and associated interpretations.

The verification has been conducted in accordance with the VCS Standard version 4.7, other valid and applicable VCS Regulatory Documents and templates (see Appendix 2 of this verification Report below), UNFCCC rules and requirements (where applicable) and associated decisions of the VCS Secretariat and the UNFCCC Executive Board (where applicable) (hereinafter also can be referred to as regulatory documents).

The verification is not meant to provide any consulting towards the client. However, stated requests for forward actions and/or corrective actions may provide input for improvement of the Project monitoring towards reductions in the GHG emissions.

1.3 Level of Assurance

The verification report is based on the VCS-PD, VCS Monitoring Report (MR), supporting evidences made available to the verifier as well as other supporting evidences made available to the VVB and information collected through performing interviews and during the on-site assessment.

Through an exhaustive process of documentary review, interviews with the project owners, site visits and sampling and surveying (where applicable), assessment on GHG Emission Reductions calculations against applicable methodology and any other regulatory document, and reporting of objective evidences, the VVB is able to reach a reasonable level of assurance on the conduction of the Verification process.

The achieved level of assurance for this verification is thus a reasonable level of assurance as per the requirements set out in the applied VCS Standard version 4.7 clause 4.1.10 (1).

VVB applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS Standard version 4.7 clause 4.1.10 (4).

Additionally, according to the VCS Validation and Verification Manual version 3.2, while all material errors, omissions and misrepresentations must be addressed for a project to receive a positive validation or verification opinion, when non-material errors have been found in the project documents, the VVB requested the project proponent to correct them and ensured that such errors have been addressed where practicable.

The verification conclusion is assured a reasonable level of assurance.

1.4 Summary Description of the Project

Project title	Fujian Jiangle Dayan 32MW Hydropower Project
VCS reference number	246
Project Proponents	Fujian Jinhu Power Co., Ltd. (Project Owner, host country, P. R. China)
Location of the project	Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China. Geographic coordinates: longitude of 26.6543° E and latitude of 117.2575° N
Project start date	Construction start date: 04/04/2003 Operation start date: 15/12/2004

Version of PD	VCS PD Version 2, dated 29/10/2009
Monitoring period	28/03/2006 to 27/03/2012
First monitoring report	Version 01, dated 25/04/2024
Final monitoring report	Version 02, dated 25/07/2024
Applied Methodology/Version	ACM0002, version 10.0
Scope/Technical Area	1/1.2 Scope 01: Energy industries (renewable-/non-renewable sources)

The project activity is a hydropower generation project located at Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China, of which the main purpose is to use hydropower for electricity generation. The total installed capacity of the project activity is 32 MW, consisting of 2 sets of hydropower turbine generators with unit capacity of 16 MW. During the fixed 10 years crediting period, the average annual power delivered to the grid by the project is expected to be 120,282 MWh. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of East China Power Grid (ECPG). It's estimated that the proposed project could achieve GHG emission reductions of 105,990 tCO₂e annually.

The project activity has been validated by TÜV SÜD Industrie Service GmbH based on the VCS PD /3/ version 2 dated 29/10/2009 and reported in the validation report No. 600500301 /4/, completed on 18/11/2009 which is available at <https://registry.verra.org/app/projectDetail/VCS/246>.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Verification was conducted using Applus+ Certification's procedures in line with the requirements specified in the VCS Standard version 4.7, CDM M&P, the latest version of the CDM Validation and Verification Standard, and relevant UNFCCC requirements and applying standard auditing techniques.

Applus+ Certification verified the implementation of the monitoring plan and the data presented in the Monitoring Report for the period in question. This involved a site visit and a desk review of the Monitoring Report. This Verification Report describes the findings of this assessment.

The Verification process has been conducted following the regulatory documents and based on the ISO 14064-3:2019 and ISO 14065:2020 using standard auditing techniques as required by the VCS Standard version 4.7.

The Verification process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this Validation (cf. VCS Standard version 4.7. Para 4.1.13).

No ICT means have been used for the conduction of this Verification process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of the Verification process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements.

This process has included: the review of data and information presented to verify their completeness; the review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements and the QA/QC procedures; an evaluation of data management; and an evaluation of the QA/QC system in place in the context of its influence on the generation and reporting of emission reductions.

No virtual sites have been included in the Verification process according to the definition in the IAF MD4: 2023 document. Instead, the documentation has been provided upon request from the VVB by other means (such as physical documents, mail, transfer applications, etc.)

Applus+ Certification has completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project's emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;

- Means of verifying reported data; and
- Compilation of the monitoring report.

Based on the above, the VVB prepared a Verification Plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the verification. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this Verification (5% as per the applied VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the statements made by the client against the principles and requirements of the applied regulatory documents during this verification.

The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:

- The selection and management of the data and information;
- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;
- Systems and processes in place to ensure the validity and accuracy of the data and information;
- The design and maintenance of the system to control the data and information;
- Systems, processes and personnel that support system in place, including activities for ensuring data quality;
- The plans for instrument maintenance and calibration where appropriate;
- The results of previous validation activities and verifications, where available and appropriate

Applus+ Certification has verified the implementation of the monitoring plan and the data presented in the Monitoring Report (MR) version 02, dated on 25/07/2024 for the monitoring period applicable to this verification process.

The process of verification involved the following relevant milestones:

- The signature of the contract for verification between the PP and the VVB;
- A kick-off meeting/communication to require the documentary material to perform an initial desk review;
- An initial desk review of the documentary evidences prepared by the Project Proponent;
- The issuance of findings coming from this initial desk review (if any);
- The conduction of an onsite audit, including interviews and video-calls in substitution of a site visit (see Sections 2.3 and 2.4 below);
- The issuance of findings from the onsite audit and communication of deviations (if any) to the Project Proponent;
- The assessment of their resolution and closure, when appropriate;
- Once all the findings have been closed, the elaboration of this Verification Report, in which such findings (if any) are described;
- The conduction of an independent Technical Review;

- Upon conclusion of the Technical Review, the VVB may perform a final quality check before the issuance of the Final Verification Report (FVR) and signature of the corresponding Verification Deed.

The process of this verification assessment has not implied any sampling plan.

The information of the assessment team is included in below:

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of assessment team has to be approved by the LGAI Technological Center S.A. (Applus+ Certification) Central Site during its Contract Review stage, ensuring that the required skills are covered by the team.

The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A).
- Technical Expert (TE).
- Technical Reviewer (TR).
- Other supporting roles such as Country Expert (CE) or Financial Expert (FE)
- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the Assessment Team as shown below:

Name	Role	SS/TA Knowledge	Financial Expertise	Host country experience	Attendance to on-site visit
Doris Dai	LA/TE	Yes (1.2)	n/a	Yes	Yes
Simon Shen	TR /TE	Yes (1.2)	n/a	Yes	n/a

Doris Dai (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ Certification (LGA Technological Center, S.A) for the GHG project assessment and auditing. She has more than 8 years of work experience in CDM/VCS project assessment and more than 3 years of working experience in GS4GG project assessment. Before she joined Applus+ Certification (LGA Technological Center, S.A), she has been working for CTI Certification as senior GHG Auditor for 3.5 years.

Simon Shen (Master's degree in Thermal Energy Engineering, Bachelor's Degree in Environmental Engineering) is an Auditor appointed by Applus+ Certification (LGA Technological Center, S.A) for the GHG project assessment, auditing and technical review. He has more than 8 years of work experience in CDM/GS4GG/VCS project assessment and review with Applus+, apart from the years of experience working as GHG Auditor and ISO 9001/14001 in TUV SUD for 3.5 years before he joined Applus+.

Mr. Simon Shen has extensive experience also as former Applus+ Shanghai CDM Technical Manager. Since 2014, Mr. Simon Shen works as an external individual in Applus+. At the time, he participated in plenty of Chinese CCER audits and enterprises carbon emissions verifications.

2.2 Document Review

The VCS monitoring report /1/ version 01 dated 25/04/2024, version 02 dated 25/07/2024 and the emission reduction calculations spreadsheet /2/, were assessed as part of the verification. In addition, the VCS PD /3/ version 2 dated 29/10/2009 in particular the baseline estimations and the monitoring plan, the VCS Validation Report /4/ dated 18/11/2009, as well as relevant documents, were reviewed. A detailed document reviewed are listed in Appendix 2 of the report.

2.3 Interviews

A physical site visit interview was made by the assessment team on 16/07/2024, The key personnel interviewed through the sit visit are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Xiao Xinghua	Director	Fujian Jinhua Power Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance.
Mr. Lin Peng	Operator	Fujian Jinhua Power Co., Ltd.	
Mr. Huang Binhao	Operator	Fujian Jinhua Power Co., Ltd.	
Ms. Yang Dongying	Villager	Dayan Village	
Mr. Xiong Jiakai	Villager	Dayan Village	
Mr. Lin Chuanfeng	Villager	Dayan Village	
Mr. Zheng Jiangzhuo	Project Manager	Beijing 3050 Technology Co.,Ltd	Data collection and ER calculation.

2.4 Site Visits

The assessment team performed the on-site verification (Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China) on 16/07/2024. The interviewed personnel including the role and organization and objective are listed in above table in the section 2.3, please refer to above for details.

2.5 Resolution of Findings

As an outcome of the verification process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the Project Proponents, or if the evidence provided to prove conformity is insufficient;
- b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the Project Proponents;
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the Project Proponents.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

All CARs and CLs raised during verification shall be resolved prior to submitting a request for issuance.

There are no CARs and CLs raised for this monitoring period for the project.

2.5.1 Forward Action Requests

None FAR was raised during the verification process. Also, there are no remaining from former validation.

2.6 Eligibility for Validation Activities

Not applicable as LGAI Technological Center, S.A. holds the accreditation for the validation and verification for projects under scope 1.

3 VALIDATION FINDINGS

3.1 Methodology Deviations

Not applicable as no deviation for methodology.

3.2 Project Description Deviations

There are 2 project description deviations for the project during this monitoring period:

1. Based on registered PD, leakage from the diesel consumed of the diesel generator has been considered, however based on ACM0002, leakage from the consumption of fossil fuel should be considered as project emissions or leakage. Therefore, such emissions from diesel consumed of the diesel generator has been removed during this monitoring period. Also related monitoring parameters are also removed from monitoring plan.
2. According to registered PD, the monitoring of Net electricity supplied to the ECPG by the project in year y involved in 4 bi-directional electricity meters (M1, M2, M3 and M4), however based on power purchase agreement and site visit, it is confirmed during this monitoring period only M1 and M2 are used for monitoring, therefore M3 and M4 are removed from the monitoring plan.

By site visit and checking related documents, it is confirmed that the deviations applied would not affect the applicability of the methodology, additionality and the appropriateness of the baseline scenario. After the deviation, the project remains in conformance with the VCS rules and the deviation is valid.

3.3 New Project Activity Instances in Grouped Projects

Not applicable as not a grouped project.

3.4 Baseline Reassessment

Not applicable as the project does not undergo baseline reassessment during the monitoring period.

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence gathering activities, evidence checked, and assessment conclusion:
Audit history	By checking registered PD /3/ and information on the VERRA website, the audit history mentioned in the MR has been confirmed to be correct
Double counting and participation under other GHG programs	Through reviewing the registered PD /3/ and validation report /4/, it was validated that the project has been registered as a VCS project with reference No. 246 which is available at https://registry.verra.org/app/projectDetail/VCS/246. Sufficient information has been provided for registered or seeking registration under any other GHG programs or rejected by another GHG program. The project does not participate in other emissions trading program by checking public information on Internet, interviewing with project owner and statement issued by project owner. During the period from 28/03/2006 to 27/03/2016 as the VCS crediting period, the project would claim only for VCUs. Also, a declaration from project proponent has been checked and in which indicated that no credits would be claimed under any other GHG programs for this monitoring period. Therefore, Applus+ Certification consider the project is eligible to participate under the VCS Program as there is no double counting for the emission reduction during any period.
No double claiming with emissions trading programs or binding emission limits	The GHG emission reductions or removals generated by the project have not become included in an emissions trading program or any other mechanism that includes GHG allowance trading. The same has been checked through declaration from the project proponents.

	The assessment team checked the REC Mechanism database of China, Chinese Emission Trading System (Chinese ETS) and Chinese Certified Emission Reduction Mechanism (CCER), and found that the project activity is not accredited / registered under REC mechanism, Chinese ETS, or CCER mechanism. Therefore, Applus+ Certification confirm no double claiming with emissions trading programs or binding emission limits.
No double claiming with other forms of environmental credit	The project has not received or sought any other form of environmental credit or has become eligible to do so since validation or previous verification. The same has been checked through declaration from the project proponents. By checking information on I-REC website /20/ and REC Mechanism database of China, it is confirmed that the project does not apply for REC of China or I-REC. Therefore, Applus+ Certification confirm no double claiming with other forms of environmental credit.
Supply chain (scope 3) emissions double claiming	The project does not involve in supply chain (scope 3) emissions double claiming as all electricity generated by the project was supplied to power grid. The project is hydropower generation project does not involve in supply chain (Scope 3).
Sustainable development contributions	By checking registered VCS PD /3/, EIA Report /28/, HR Records /16/, China's National Plan on Implementation of the 2030 Agenda for Sustainable Development and 17 SDGs defined by UNDP, interviewing with stakeholders during site visit, the verification team confirmed that the project activity would contribute sustainable development in the region and China's Sustainable Development Goals (SDG). The specific sustainable development goals that the project activity could achieve have been listed as follows: SDG 8 Decent works and Economic Growth The project generated 34 decent working opportunities for local women and men, 17 for women and 17 for men and help them improve the incomes, and contributes to the decent work and economic growth. SDG 7 Affordable and Clean Energy

	The project has supplied 634,403.386 MWh in this monitoring period that otherwise have been produced by thermal power generation projects, and contributes to Chinese renewable energy target. SDG 13 Climate Action The emission reduction of the project is 563,855 tCO₂e in this monitoring period, and contributes to Chinese Climate change target. Moreover, by checking public information, staff roaster of project /16/ and site visit, information mentioned in the Table of Sustainable Development Contributions in the MR has been confirmed to be correct.
Additional information relevant to the project	No commercially sensitive information has been excluded from the public version of the PD or MR. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this verification.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	By checking registered PD /3/, and site visit, the stakeholder involved in the project has been correctly identified in the MR and in line with the same in the PD as below: The project is located in upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China, local residents in Dayan Village are considered as stakeholders. Since the project is governed by local government, local government as Jiangle County government are considered as stakeholders. Also, the staffs of the project are considered as stakeholders. The assessment team considers the stakeholder identification of the project activity as reasonable.

Legal or customary tenure/access rights	<i>By checking project approval of the project /21/, it is confirmed that the project owner has the legal right to use hydro resource for power generation.</i> <i>As per FSR and its approval, the implementation of the project would result in the submerge of 127 Mu cultivated land and 23.5 Mu Forest land, which is collective owned land and held by local communities, which has been compensated according to relevant national compensation regulation, the legal or customary tenure/access rights to was protected as per the national law/regulations. The same has been confirmed by checking compensation agreement /19/ and site visit interview.</i> <i>By site visit and checking documents, it is confirmed the project does not involved in land issues.</i> <i>Therefore, it was verified the project activity does not violate any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i>
Stakeholder diversity and changes over time	<i>Consider there are no change for the project since the validation and the project is implemented as the design mentioned in the PD, the stakeholder diversity and changes over time is highly unlikely and this is the same as actual scenario as no stakeholder diversity and changes occurs.</i> <i>And during this monitoring period, there are no change for the stakeholders.</i>
Expected changes in well-being	<i>Following expected changes in well-being have been identified and confirmed to be correct through the site visit:</i> <i>Economic well-being</i> <i>The project has supplied renewable energy to local area, reduce the electricity shortage in the coastal area of Fujian province and contribute to the economic growth in the area.</i> <i>Social well-being</i> <i>The project generated 34 working opportunities for local residents, and help them improve the incomes and the likelihood.</i> <i>Environmental well-being</i>

	<i>The project has supplied renewable energy to local area that otherwise have been produced by thermal power generation projects, the implementation of the project would reduce the pollutants of SO₂, NO_x, particulate matter and GHG emission.</i>
Location of stakeholders	<i>By site visit, it is confirmed that the location of stakeholders is in Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China.</i>
Location of resources	<i>By site visit, it is confirmed that the location of resources is in Upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian Province, P. R. China.</i>

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Ongoing consultation	<i>By site visit interview, it is confirmed that during the monitoring period, a grievance expression book /22/ was put in the office of the project, which is used to collect stakeholders' views about the project. Project Proponent checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information.</i> <i>Meanwhile, during this monitoring period, stakeholder meetings have been held.</i>
Date(s) of stakeholder consultation	<i>By checking questionnaires /23/, it is confirmed that a local stakeholder consultation meeting was held on 01/09/2004 by the project owner. Before the stakeholder consultation meeting, the project information with contact information and the stakeholder meeting notice were published on the bulletins at and nearby the project site to invite Local stakeholders.</i> <i>During the meeting, the project information and the continuous input / grievance mechanism have been introduced to the local stakeholders. The project information with contact information has also been posted on the bulletins at and nearby the project site.</i> <i>During the meeting, a survey was carried out on the local residents and comments received from the survey are summarized as follows.</i>

	The survey was conducted through distributing and collecting responses to a questionnaire. In total 30 out of 30 questionnaires were returned with a 100% response rate. By checking meeting minutes /24/ and site visit, it is confirmed that during this monitoring period, 6 meetings were organized at project site on 09/09/2006, 02/09/2007, 04/09/2008, 05/09/2009, 04/09/2010 and 02/09/2011 to collect comments from all stakeholders, such as representatives of local residents, local governments, and the project owner and so on. The project owner appointed one person to make a meeting minute.
Communication of monitored results	By checking the questionnaires, it is confirmed that the local stakeholder has no negative comments for the construction of the project activity. By checking meeting minutes and site visit, it is confirmed during this monitoring period, 6 stakeholder consultation meetings were held. At each meeting, the plant manager informed the stakeholders about the recent operation of the project. Questions raised by stakeholders were answered on site.
Consultation records	By checking the questionnaires, questionnaires have been filled out during the stakeholder meetings before the implementation of the project. By checking grievance expression book, meeting minutes and site visit, it is confirmed that each time a stakeholder consultation meeting is held, one person is designated to record the meeting. The questions and suggestions raised by stakeholders at the meeting were recorded. All the information has been recorded as meeting minutes. A grievance expression book was put in the office of the project, which is used to collect stakeholders' views about the project. Project Proponent checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information.
Stakeholder input	By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that there were no negative inputs received during both construction and operation phase.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	<i>By checking Project approval /25/ and EIA approval /26/ issued by government, it is confirmed the project has been approved by Chinese government.</i> <i>By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that local stakeholders held full right for FPIC.</i> <i>Therefore, it was verified the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i> <i>And the project does not exacerbate nor influence the outcomes of unresolved conflicts.</i>
Outcome of FPIC discussion	<i>As mentioned above, the project has been approved by Chinese government and local management institute. And local stakeholders have no negative comments for the construction and operation of the project.</i> <i>The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.</i> <i>The assessment team confirmed that the local stakeholders held full right for FPIC. And the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i>

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	<i>Not applicable as no negative comments have been received for the project during both construction and operation phase of the project. Also, the assessment team confirm the procedure is easily accessible to stakeholders for ongoing consultation as described as above.</i>

Grievance redress procedure	<i>Not applicable as no negative comments have been received for the project during both construction and operation phase of the project. Also, the assessment team confirm the procedure is easily accessible to stakeholders for ongoing consultation as described as above.</i>
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4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
<i>No negative comments received from public channel.</i>	<i>Not applicable as no negative comments received from public channel.</i>	<i>By checking questionnaires, grievance expression book, meeting minutes, VERRA website and site visit, it is confirmed that there were no negative inputs received during both construction and operation phase.</i>

4.2.6 Risks to Local Stakeholders and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.</i> <i>Also, during the site visit, no risks have been identified to stakeholder participation.</i> <i>Therefore, it was verified no risks have been identified to stakeholder participation.</i>
Working conditions	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.</i>

	Also, during the site visit, it is confirmed that the staff of the project is working under indoor environment and safety protection equipment have been provided in case for on-site monitoring and device maintenance. No risks identified for working conditions.
Safety of women and girls	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws /27/ set up by Chinese government in aspect of environment, health and safety. Moreover, China has published "Women's Rights Protection Law" to provide support and protection to women and girls. In order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. Also, China published "Women's Rights Protection Law". Therefore, the Project will not expose women and girls to further risks or hazards. By checking HR Records and Staff Roaster, it was verified no girls have been recruited, also equal payment has been made to women and men. Also, during the site visit, it is confirmed that no risks has been identified to safety of women and girls.
Safety of minority and marginalized groups, including children	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of minor protection. The country has strict prohibition for child labour. Thus, project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview. Besides, in order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. By checking HR Records and Staff Roaster and site interview, the assessment team was able to confirm that no Minors under the age of 16 were employed. Also, during the site visit, it is confirmed that no minors have been hired by the project owner. No child labor detected for the project. No risks

	identified for safety of minority and marginalized groups, including children as not applicable for the project.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment. During the operation period, all the mitigation measures proposed by EIA report /21/ will be implemented and the following key aspects will be addressed: Impact on social economics The loss of submerged properties of the project is small. The natural condition of the Dayan Project is an advantage for its construction; the economic benefit is quite significant. The project will contribute to rural electrification development and to alleviate electricity shortage in Jiangle County. It is an ideal site for construction of hydropower station. Building up of the project complies with the Jinxi River cascade hydropower station development, which is to promote the development of hydro powers. The project also meets the requirement of The Comprehensive Plan and Design of Jinxi River, which is a guideline from local government to mark out the development of the basin. Impact on quality of surface water Waste water discharged during the construction period mostly comes from construction process and daily life. The construction company will construct deposit pool, sedimentation pool and oil blocking pool to dispose the waste water from construction; cesspool will be established to mitigate the direct pollution of waste water from daily life. Eutrophication impact Based on the research of reservoir eutrophication impact both in the country and overseas, Total Phosphorous (P) is selected as the assessment indicator of eutrophication impact.

Phosphorous was not detected before the reservoir construction. Phosphorous may come from daily life, soil erosion and emerged forest land, plantation and plants. Using an assessment model (Dillon Model) to calculate the P content in the reservoir after the project begin operation, the result is 0.013 mg/L. According to the standard of P level listed below, Dayan reservoir will remain poor nutrition so that eutrophication will not happen.

Environmental impact during construction

The troubling and organic matters are increased in the lower reach of the river, the noise standard of living and office area is higher than the normal. There are dust produced in the construction area, however, the impact can be reduced by installing the dust shields for the operation machines. The project owner shall also plant trees and grasses around the project site to reduce noise and dust.

Impact on soil erosion

The project occupies land 127 mu, includes forest and plantation. The soil erosion affected area includes 23.5 mu forest, 114 mu plantation.

There is solid waste from the project construction, vegetation recovery and proper handling the solid waste are primary concern for preventing the soil erosion.

In conclusion of the EIA report, the project does not have any negative significant impacts on the environment.

4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena. By site visit, it is confirmed no discrimination and sexual harassment detected.

	<i>The project does not create any direct or indirect impacts on gender equality and/or the situation of women:</i> <i>a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.</i> <i>b. The Project will not restrict women's rights or access regarding natural resources.</i> <i>c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.</i>
Management experience	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government which means the government recognize the experience of the project owner for project construction and operation. Moreover, the project owner of the project has developed several landfill gas projects and therefore has extensive development and management experience.</i>
Gender equity in labor and work	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.</i> <i>By site visit, it is confirmed no violation for Gender equity in labor and work detected.</i> <i>The project, complying with the "Women's Rights Protection Law" and "Labor Contract Law" does not discriminate the local community on basis of gender or caste or religion and therefore equally serve to all.</i>

	By checking HR Records and interviewing staff from the project owner, it was verified the Project design neither increase women's workload nor prevent them from engaging in other activities. The Project has equal opportunity for women and men to contribute both in volunteer and working positions. The project has HR policy and same is followed equally. The project owner has a specified HR policy that considers participation by both men and women. Besides, China also ratified relevant ILO core conventions on equality, namely Equal Remuneration Convention (Convention No 100). Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.
Human trafficking, forced labor, and child labor	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena. Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China. China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee. China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview. By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.

4.2.7.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Human rights	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. Therefore, it is confirmed that the project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law" to provide support and benefits to women.
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4.2.7.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. Hence not applicable for the project.

4.2.7.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Disputes over rights to territories and resources	Not applicable as no IPs', LCs', and customary rights holders' property rights was identified.
Respect for property rights	Not applicable as no IPs', LCs', and customary rights holders' property rights was identified.

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	<i>Not applicable as no property rights involved for the project.</i>
Benefit sharing during the monitoring period	<i>Not applicable as no property rights involved for the project.</i>

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	<i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Soil degradation and soil erosion	<i>The Project's purpose is to supply electricity from hydropower. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Water consumption and stress	<i>The Project's purpose is to supply electricity from hydropower. Therefore, the impact of the Project on water consumption and stress is negligible.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Usage of fertilizers	<i>The project is a hydropower generation project. Therefore, the Project does not involve the application of fertilizers.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Species or habitat	<i>By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.</i>
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4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
<i>Not applicable</i>	<i>Not applicable as no rare, threatened, or endangered species involved.</i>

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
<i>Not applicable</i>	<i>Not applicable as no rare, threatened, or endangered species involved.</i>

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	<i>Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.</i>

4.3 Accuracy of Reduction and Removal Calculations

By means of on-site visit, the assessment team confirms that all physical features of the proposed project activity in the registered VCS PD /3/ are in place and the PP has operated the project as per registered VCS PD /3/. The installed capacity of the project is 32 MW, involves the installation of 2 sets of hydro turbine generators with a unit capacity of 16 MW. The electricity generated is transmitted to the local Power Grid via a newly built transformer station, which was then exported to the ECPG. The project activity was expected to supply 120,282 MWh of electricity to the grid annually. The construction of the project started on 04/04/2003, the project has been put into fully operation on 25/02/2005 verified by checking information on the VCS website and site visit. There are no changes on the key equipment and technology since the validation of the project. No special event which would affect the monitoring of the project has been observed during the monitoring period. The project was under smooth operation except regular repair and maintenance confirmed by checking operation log.

The technical parameters have been verified with the nameplates /9 / as below:

Turbine (s)		Generator (s)	
Number of Units	2	Number of Units	2

Turbine Type	GZD278A-WP-470	Type	SFWG16-52/1340
Rated capacity	16.667	Rated power	16 MW
Rated speed	115.4 r/min	Rated voltage	6 kV
Design head	9.5		
Single-unit Discharge capacity	191.5 m ³ /s		
Manufacturer	Dongfang Electric Co.,Ltd	Manufacturer	Dongfang Electric Co.,Ltd
Lifetime of the equipment (years)	30	Lifetime of the equipment (years)	30

The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period.

All required equipments and procedures are available and implemented in an appropriate manner.

All necessary monitoring instruments are installed. All required instruments including standby and operating procedures for the same have been implemented in an appropriate manner.

The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD /3/. All parameters were monitored and determined as per the monitoring plan which is listed in below table:

Data / Parameter:	Electricity Exports (EG _{EX})
Data unit:	MWh
Description:	Annual electricity supplied to the ECPG by the project
Purpose of the data:	Calculation of baseline emissions
Parameter value:	634,403.386
Source of data used:	Annual electricity supplied to the ECPG by the project (Electricity Exports (EG _{EX})) are all sourced from Meter Reading Record (MRRs) /10/ issued by the project owner, Electricity Transaction Notes (ETNs) /11/ issued by power grid company and Statement issued by power grid company /18/ covering monitoring period.
Information flow:	2 sets of bidirectional electricity meters (one set as main (M1) and another as backup (M2)) installed located at the high-voltage side of the on-site

	transformer station were measured continuously, recorded monthly and archived electronically. At 24:00 hr of 27th of each month, the staff from project owner and power grid company will record 2 sets of bidirectional electricity meters' readings and form Meter Reading Records (MRRs) and then transcribes the data into Electricity Transaction Notes (ETNs) by power grid company. For data of 31/12/2006, 31/12/2007, 31/12/2008, 31/12/2009 31/12/2010 and 31/12/2011 are determined by MRRs and Statement issued by power grid company. The data for MRRs, ETNs and Statement issued by power grid company have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation.												
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 2 sets of bidirectional electricity meters (one set as main (M1) and another as backup (M2)) installed located at the high-voltage side of the on-site transformer station during the monitoring period verified by site visit. See below for the information of 2 sets of bidirectional electricity meters verified by site visit and checking calibration certificates /12/: <table><tr><th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>M1</td><td>DTZ341</td><td>EP11143408</td><td>0.5s</td></tr><tr><td>M2</td><td>DTZ341</td><td>EP11143397</td><td>0.5s</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.	Meter	Type	Serial Number	Accuracy	M1	DTZ341	EP11143408	0.5s	M2	DTZ341	EP11143397	0.5s
Meter	Type	Serial Number	Accuracy										
M1	DTZ341	EP11143408	0.5s										
M2	DTZ341	EP11143397	0.5s										
Calibration:	During this monitoring period, no calibration has been made, but a calibration was made after the end of monitoring period as below: <table><tr><th>Meter</th><th>Calibration date</th><th>Valid until</th></tr><tr><td>M1</td><td>03/11/2022</td><td>02/11/2023</td></tr><tr><td>M2</td><td>03/11/2022</td><td>02/11/2023</td></tr></table> the calibration was conducted by accredited third parties which is State Grid Fujian Electric Power Co., LTD was accredited by Administration for Market Regulation Fujian Province /13/. Considering, no calibration was made during this monitoring period, but calibration was made after, therefore, a max permissible error factor (0.5%) has been considered in the ER calculation which met the requirement,	Meter	Calibration date	Valid until	M1	03/11/2022	02/11/2023	M2	03/11/2022	02/11/2023			
Meter	Calibration date	Valid until											
M1	03/11/2022	02/11/2023											
M2	03/11/2022	02/11/2023											
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.												

Means of verification:	Data of the parameter was verified by checking MRRs. All data is in line with MRRs; Information flow was verified by checking MRRs, ETNs and Statement issued by power grid company, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate and Accreditation certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.
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Data / Parameter:	Electricity Imports (EG _{im})
Data unit:	MWh
Description:	Annual electricity imported to the project from the ECPG
Purpose of the data:	Calculation of baseline emissions
Parameter value:	0
Source of data used:	Annual electricity imported to the project from the ECPG (Electricity Imports (EG _{im})) are all sourced from Meter Reading Record (MRRs) /10/ issued by the project owner, Electricity Transaction Notes (ETNs) /11/ issued by power grid company and Statement issued by power grid company /18/ covering monitoring period.
Information flow:	2 sets of bidirectional electricity meters (one set as main (M1) and another as backup (M2)) installed located at the high-voltage side of the on-site transformer station were measured continuously, recorded monthly and archived electronically. At 24:00 hr of 27th of each month, the staff from project owner and power grid company will record 2 sets of bidirectional electricity meters' readings and form Meter Reading Records (MRRs) and then transcribes the data into Electricity Transaction Notes (ETNs) by power grid company. For data of 31/12/2006, 31/12/2007, 31/12/2008, 31/12/2009, 31/12/2010 and 31/12/2011 are determined by MRRs and Statement issued by power grid company. The data for MRRs, ETNs and Statement issued by power grid company have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation.

Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 2 sets of bidirectional electricity meters (one set as main (M1) and another as backup (M2)) installed located at the high-voltage side of the on-site transformer station during the monitoring period verified by site visit. See below for the information of 2 sets of bidirectional electricity meters verified by site visit and checking calibration certificates /12/: <table><tr><th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>M1</td><td>DTZ341</td><td>EP11143408</td><td>0.5s</td></tr><tr><td>M2</td><td>DTZ341</td><td>EP11143397</td><td>0.5s</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.	Meter	Type	Serial Number	Accuracy	M1	DTZ341	EP11143408	0.5s	M2	DTZ341	EP11143397	0.5s
Meter	Type	Serial Number	Accuracy										
M1	DTZ341	EP11143408	0.5s										
M2	DTZ341	EP11143397	0.5s										
Calibration:	During this monitoring period, no calibration has been made, but a calibration was made after the end of monitoring period as below: <table><tr><th>Meter</th><th>Calibration date</th><th>Valid until</th></tr><tr><td>M1</td><td>03/11/2022</td><td>02/11/2023</td></tr><tr><td>M2</td><td>03/11/2022</td><td>02/11/2023</td></tr></table> the calibration was conducted by accredited third parties which is State Grid Fujian Electric Power Co., LTD was accredited by Administration for Market Regulation Fujian Province /13/. Considering, no calibration was made during this monitoring period, but calibration was made after, therefore, a max permissible error factor (0.5%) has been considered in the ER calculation which met the requirement,	Meter	Calibration date	Valid until	M1	03/11/2022	02/11/2023	M2	03/11/2022	02/11/2023			
Meter	Calibration date	Valid until											
M1	03/11/2022	02/11/2023											
M2	03/11/2022	02/11/2023											
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.												
Means of verification:	Data of the parameter was verified by checking MRRs. All data is in line with MRRs; Information flow was verified by checking MRRs, ETNs and Statement issued by power grid company, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate and Accreditation certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.												

Data / Parameter:	Net Electricity Exports (EG _y)
Data unit:	MWh
Description:	Net electricity supplied to the ECPG by the project in year y
Purpose of the data:	Calculation of baseline emissions
Parameter value:	634,403.386
Source of data used:	Net Electricity Exports (EG_y) is the difference between Electricity Exports (EG_{Ex}) and Electricity Imports (EG_{im}) as below formula: $EG_y = EG_{Ex} - EG_{im}$ For both Electricity Exports (EG_{Ex}) and Electricity Imports (EG_{im}), please refer to above report.
Information flow:	Net Electricity Exports (EG_y) is the difference between Electricity Exports (EG_{Ex}) and Electricity Imports (EG_{im}) as below formula: $EG_y = EG_{Ex} - EG_{im}$ For both Electricity Exports (EG_{Ex}) and Electricity Imports (EG_{im}), please refer to above report.
Monitoring method, frequency and equipments:	Not applicable
Calibration:	Not applicable
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Not applicable

Data / Parameter:	A _{PJ}
Data unit:	m ²
Description:	Area of reservoir at full reservoir level
Purpose of the data:	Calculation of project emissions
Parameter value:	2006: 1,871,000 2007: 1,871,000 2008: 1,871,000 2009: 1,871,000 2010: 1,871,000

	2011: 1,871,000 2012: 1,871,000
Source of data used:	Area of reservoir at full reservoir level (A_{PJ}) is determined by checking Statement issued by Fujian Provincial Investigation & Design Institute of Water Conservancy & Hydro Power /28/.
Information flow:	Area of reservoir at full reservoir level (A_{PJ}) is determined by checking Statement issued by Fujian Provincial Investigation & Design Institute of Water Conservancy & Hydro Power.
Monitoring method, frequency and equipments:	Not applicable
Calibration:	Not applicable
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Area of reservoir at full reservoir level (A_{PJ}) is determined by checking Statement issued by Fujian Provincial Investigation & Design Institute of Water Conservancy & Hydro Power.

Data / Parameter:	Cap_{PJ}
Data unit:	W
Description:	Installed capacity of the hydropower plant after the implementation of the project activity
Purpose of the data:	Calculation of project emissions
Parameter value:	32,000,000
Source of data used:	Installed capacity of the hydropower plant after the implementation of the project activity (Cap_{PJ}) is determined by checking nameplate of the equipment on site.
Information flow:	Installed capacity of the hydropower plant after the implementation of the project activity (Cap_{PJ}) is determined by checking nameplate of the equipment on site.
Monitoring method, frequency and equipments:	Not applicable
Calibration:	Not applicable

QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Installed capacity of the hydropower plant after the implementation of the project activity (Cap_{PJ}) is determined by checking nameplate of the equipment on site.

Parameters available at validation stage:

Below data has been verified against the data sources and the PD, for the rest of parameters are not used in the calculation of ER in this monitoring period.

Parameter title	Description	Data	Source
$EF_{grid,CM,y}$	The baseline combined margin CO_2 emission factor for grid connected power generation in year y	0.8888	2008 Baseline Emission Factors for Regional Power Grids in China published by China DNA /14/

In conclusion, the assessment team confirmed GHG emission reductions and removals have been quantified correctly in accordance with the project description and applied methodology.

4.4 Quality of Evidence to Determine Reductions and Removals

The monitoring has been carried out in accordance with VCS PD /3/ dated 29/10/2009.

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the MR Report and ER Calculation Spreadsheet /2/ and are consistent with the applied methodology ACM0002 version 10.0 and the monitoring plan. The parameters are complete in this monitoring period.

By checking the original record, crosscheck with the supporting evidence issued from other party than project owner, checking calibration related documents and interview with project owner through the site visit, the VVB is able to confirm there are no transposition errors between data sets. All data are consistent in all data sets.

After verifying the reported figures with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation Spreadsheet and the MR Report. The verification process for the same has been clearly described above in section 4.3 of the report.

4.5 Non-Permanence Risk Analysis

Not applicable as a landfill gas project.

5 VERIFICATION OPINION

5.1 Verification Summary

LGA Technological Center S.A. (hereinafter referred to as Applus+ Certification) has been commissioned by Beijing 3050 Technology Co.,Ltd to perform the verification of greenhouse gas emission reductions of the project activity “Fujian Jiangle Dayan 32MW Hydropower Project” (VCS Ref. No. 246).

The management of Fujian Jinhu Power Co., Ltd. is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the VCS PD /3/ dated 29/10/2009.

Our verification approach was based on the requirements as defined under the applicable VCS standards, relevant UNFCCC requirements and ISO 14064-3: 2019. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered PD;
- the monitoring plan in the registered PD is as per the applied methodology;
- the monitoring complies with the registered PD;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

In our opinion, the GHG emission reductions for “Fujian Jiangle Dayan 32MW Hydropower Project” during the monitoring period 28/03/2006 to 27/03/2012 as reported in Monitoring Report, prepared on the basis of the project’s Monitoring Plan are fairly stated.

5.2 Verification Conclusion

Our verification approach was based on the requirements as defined under the applicable VCS standards, relevant UNFCCC requirements and ISO 14064-3: 2019. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The achieved level of assurance for this verification is thus a reasonable level of assurance as per the requirements set out in the applied VCS Standard version 4.7 clause 4.1.10 (1). Based on the information we have seen and evaluated, we confirm the following statement:

Verification period: From 28/03/2006 to 27/03/2012

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Removal VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
28/03/2006 to 31/12/2006	87,183	0	0	87,183	0	87,183
01/01/2007 to 31/12/2007	98,245	0	0	98,245	0	98,245
01/01/2008 to 31/12/2008	99,391	0	0	99,391	0	99,391
01/01/2009 to 31/12/2009	70,788	0	0	70,788	0	70,788
01/01/2010 to 31/12/2010	113,106	0	0	113,106	0	113,106
01/01/2011 to 31/12/2011	70,595	0	0	70,595	0	70,595
01/01/2012 to 27/03/2012	24,547	0	0	24,547	0	24,547
Total	563,855	0	0	563,855	0	563,855

5.3 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
28/03/2006 to 31/12/2006	87,729	87,183	-0.62%	Acceptable as conservativeness, also consider this is under normal variation
01/01/2007 to 31/12/2007	98,629	98,245	-0.39%	Acceptable as conservativeness, also consider this is under normal variation
01/01/2008 to 31/12/2008	99,922	99,391	-0.53%	Acceptable as conservativeness, also

				consider this is under normal variation
01/01/2009 to 31/12/2009	106,906	70,788	-33.78%	Acceptable as conservativeness, also by checking 2009 Fujian Provincial Environmental Status communique, it is confirmed that the water flow is lower water flow year, therefore the power generation is much lower than the estimated average which lead to lower emission reduction.
01/01/2010 to 31/12/2010	106,906	113,106	5.80%	Acceptable as normal variation.
01/01/2011 to 31/12/2011	106,906	70,595	-33.97%	Acceptable as conservativeness, also by checking 2011 Fujian Provincial Environmental Status communique, it is confirmed that the water flow is lower water flow year, therefore the power generation is much lower than the estimated average which lead to lower emission reduction.
01/01/2012 to 27/03/2012	25,481	24,547	-3.67%	Acceptable as conservativeness, also consider this is under normal variation
Total	636,520	563,855	-11.42%	Acceptable as conservativeness, for certain year by checking information from local government, the decreasing is due to lower water flow.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

<i>Section</i>	<i>Information</i>	<i>Justification</i>	<i>Assessment method and conclusion</i>
<u>Not applicable</u>	<u>Not applicable</u>	<u>Not applicable</u>	<u>Not applicable</u>

APPENDIX 2: REFERENCE LIST

1. Monitoring report, Version 01, dated 25/04/2024; Version 02, dated 25/07/2024
2. ER calculation spreadsheet
3. VCS PD, version 2, dated 29/10/2009
4. VCS Validation report, No. 600500301, version 01.2, completed by TÜV SÜD Industrie Service GmbH
5. VCS standard version 4.7, dated on 16/04/2024
6. Statement issued by project owner
7. Approved methodology ACM0002: Landfill methane recovery, version 10.0
8. CDM Monitoring procedure
9. Nameplate of the equipment
10. Meter Reading Record
11. Electricity Transaction Notes covering the monitoring period
12. Calibration certificates of electricity meter M1 and M2 issued by Metrology Center of EPH
13. Accreditation certificates for Electric Power Research Institute of Guizhou Power Grid Co., LTD issued by Administration for Market Regulation Fujian Province
14. 2008 Baseline Emission Factors for Regional Power Grids in China published by China DNA
15. IPCC

- 16 Staff roster of project
- 17 Power Purchase Agreement
- 18 Statement issued by power grid company
- 19 Compensation Agreement
- 20 I-REC website: <https://www.trackingstandard.org/>
- 21 EIA report
- 22 Grievance expression book
- 23 Questionnaires
- 24 Meeting minutes
- 25 Project approval
- 26 EIA approval
- 27 Regulation and laws:
 - “Women’s Rights Protection Law”
 - “Minor Protection Law”

“Labor Law”

“Criminal Law”

- 28. Statement issued by Fujian Provincial Investigation & Design Institute of Water Conservancy & Hydro Power
- 29 2009 Fujian Provincial Environmental Status communique
- 2011 Fujian Provincial Environmental Status communique